

# Wig Sales & Profit Analytics

## Data Cleaning Process

### Purpose

The raw wig-sales dataset was intentionally created with common data-quality issues. The objective was to make the data consistent, usable and suitable for analysis while documenting important assumptions.

### Cleaning Steps

| Issue / Task       | Treatment Applied                                                      | Excel Technique            |
|--------------------|------------------------------------------------------------------------|----------------------------|
| Customer names     | Standardized capitalization and removed unnecessary spaces.            | PROPER + TRIM              |
| Wig names          | Standardized text formatting and corrected inconsistencies.            | PROPER / standardization   |
| Dates              | Converted entries to a consistent short-date format.                   | Date formatting            |
| Missing Quantity   | Estimated missing values using the median of available quantities.     | MEDIAN + replacement       |
| Duplicates         | Checked for duplicate records and removed duplicates where applicable. | Remove Duplicates          |
| Currency / numbers | Standardized presentation of price fields.                             | Number/Currency formatting |
| Calculated fields  | Created Revenue, Total Cost, Profit and Profit Margin.                 | Excel formulas             |

### Calculated Fields

| Field         | Formula / Logic          | Purpose                                     |
|---------------|--------------------------|---------------------------------------------|
| Revenue       | Quantity × Selling Price | Measures sales value generated.             |
| Total Cost    | Quantity × Cost Price    | Measures product cost.                      |
| Profit        | Revenue – Total Cost     | Measures profit generated.                  |
| Profit Margin | Profit ÷ Revenue         | Measures profit as a percentage of revenue. |

### Quality Check

After cleaning, the dataset was reviewed for consistent text, dates and numerical fields, and to ensure the calculated fields were suitable for PivotTable analysis and dashboard visualization.

### Missing-Value Assumption

Median imputation was used for missing Quantity values. This is an analytical assumption, not a recovery of the original records. In a real business setting, missing quantities should ideally be verified against source documents or the original sales system.